Kaalchakra infotech
sharing technology

PHP Lectureflow

Module-1) SE - Database Management – SQL & PL/SQL	

- Session 1: Introduction to Databases Trainer Coverage: What is a database, DBMS, RDBMS (with examples). Difference between MySQL, PostgreSQL, Oracle, SQLite. Tables, rows, columns, and relationships. Demo: Show data stored in MySQL Workbench.
- Session 2: SQL Basics & Commands Trainer Coverage: SQL syntax rules. DDL commands: CREATE, ALTER, DROP. Creating basic users and expenses tables. Demo: Execute commands in MySQL CLI / Workbench.
- Session 3: DML Operations Trainer Coverage: Insert, Update, Delete records. WHERE and LIMIT usage. Example: Insert 5 sample users and edit their data.
- Session 4: Data Retrieval with SELECT Trainer Coverage: SELECT queries, filtering, sorting, aliases. Aggregate functions (SUM, AVG, COUNT). Demo: Generate “total expenses” by category.
- Session 5: Joins and Relationships Trainer Coverage: Foreign keys and primary keys. Inner Join, Left Join, Right Join. Demo: Link users with their transactions table.
- Session 6: Grouping and Subqueries Trainer Coverage: GROUP BY, HAVING. Subqueries and nested queries. Example: “Show users whose average daily spend > ?500.”
- Session 7: Constraints & Views Trainer Coverage: Constraints: NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY. Creating and querying VIEWS. Example: Create “ActiveUserView” showing users with >5 transactions.
- Session 8: Introduction to PL/SQL Trainer Coverage: PL/SQL block structure (DECLARE, BEGIN, EXCEPTION, END). Variables and cursors. Control structures (IF, LOOP). Demo: Simple stored procedure to calculate user’s total spend.
- Session 9: Transactions & Error Handling Trainer Coverage: COMMIT, ROLLBACK, SAVEPOINT. Triggers: use case “Auto update user’s balance after expense entry.” AI Integration: Use ChatGPT to simulate test triggers and rollback scenarios.
- Mini Project 3 – “Expense Tracker DB” Description: Design the backend for a Personal Expense Tracker App. Create users, transactions, and categories tables. Write SQL scripts for CRUD operations. Add a stored procedure for monthly summary generation. Trainer Deliverables: Students should submit .sql file with queries. Test queries via MySQL Workbench live.

Module-2) SE- Introduction to Web Technologies - HTML and CSS	

- Session 1: Introduction to Web Design Coverage: How browsers render HTML + CSS. Difference between front-end & back-end. Developer tools overview (Chrome DevTools). Demo: Inspect elements on Instagram home page. Depth: Conceptual understanding of DOM structure.
- Session 2: HTML Page Structure Coverage: HTML boilerplate (<!DOCTYPE>, <html>, <head>, <body>). Tags: Headings, paragraphs, lists, anchors, images, tables. Demo: Create personal “About Me” page.
- Session 3: Forms & Inputs Coverage: Form elements (input, select, textarea), labels, validation attributes. Demo: Registration form for Social Bio Link App. Depth: Show POST request preview in DevTools Network tab.

- Session 4: Semantic HTML 5 Coverage: Semantic tags (header, nav, article, footer). Demo: Rebuild a YouTube-style page layout.
- Session 5: Introduction to CSS 3 Coverage: Inline, Internal, External styles; Selectors, Colors, Font families. Demo: Color-theme switch between light and dark mode.
- Session 6: Box Model & Positioning Coverage: Margin, Padding, Border, Display properties, Position (relative, absolute, fixed). Demo: Create a profile-card layout.
- Session 7: Flexbox & Grid Coverage: CSS Flexbox (container, justify, align) and Grid (template columns & rows). Demo: Responsive gallery of social icons.
- Session 8: Responsive Design Coverage: Media queries, Viewport, Breakpoints. Demo: Make the profile card mobile-friendly.
- Session 9: Bootstrap Basics Coverage: Bootstrap Grid System, Containers, Rows, Columns. Demo: 2-column layout with profile image and link list.
- Session 10: Bootstrap Components Coverage: Navbar, Buttons, Cards, Carousel, Forms. Demo: Navbar for Social Bio Link App.
- Session 11: Bootstrap Utilities & Customization Coverage: Spacing, Typography, Colors, Custom CSS overrides. Demo: Theme switcher using Bootstrap classes.
- Session 12: Integration Review Coverage: Integrate HTML + CSS + Bootstrap for a cohesive layout. Demo: Static mock-up of Bio Link landing page.
- Mini Project 6 – “Social Bio Link Page – UI Design” Objective: Build a personal bio link page (links to Instagram, LinkedIn, Spotify, etc.) with responsive design. Trainer Notes: Guide students to implement a form that accepts link inputs and shows preview cards.

Module 3) PHP - Core PHP

- Session 1 – PHP Introduction & Environment SetupTopics to Cover:What is server-side scripting.PHP history and its role in modern stacks.Installing XAMPP & folder structure (htdocs).Writing first script: hello.php.Demo:Run PHP file in browser via localhost.Trainer Focus:Explain server–browser request lifecycle visually.
- Session 2 – PHP Syntax, Variables & Data Types Topics to Cover: Declaring variables ($), case sensitivity. Data types: String, Integer, Float, Boolean, Array, Object, Null. Type casting and dynamic typing. Demo: Display variable types using var_dump(). Depth: Compare PHP’s dynamic typing vs Java’s static typing.
- Session 3 – Operators & Expressions Topics to Cover: Arithmetic, Assignment, Comparison, Logical, Concatenation. Operator precedence. Demo: Build a simple “Bill Calculator.”
- Session 4 – Conditional Statements Topics to Cover: if, else, elseif, switch. Nested conditions and shorthand syntax (?:). Demo: Create grading system based on user score input.
- Session 5 – Loops Topics to Cover: while, do-while, for, foreach loops. Nested loops and break/continue. Demo: Display multiplication tables (1–10).
- Session 6 – Functions Topics to Cover: Defining and invoking functions. Default parameters and return types. Variable scope and global keyword. Demo: Create reusable function calculateDiscount(). Depth: Explain DRY principle (Don’t Repeat Yourself).
- Session 7 – Arrays Topics to Cover: Indexed, Associative, Multidimensional arrays. Looping through arrays with foreach. Array functions: array_push, sort, count, in_array. Demo: Store and display student marks in an array table.

- Session 8 – Strings & String Functions Topics to Cover: Concatenation, interpolation, strlen, substr, str_replace, explode, implode. Escaping characters and trimming whitespace. Demo: Build “Username Formatter” utility.
- Session 9 – Working with Forms Topics to Cover: HTML forms with GET & POST methods. Capturing user input using $_GET and $_POST. Sanitizing input. Demo: Create “Contact Us” form capturing user info. Depth: Explain form submission lifecycle.
- Session 10 – File Handling Topics to Cover: Opening, reading, writing files (fopen, fwrite, fread, fclose). Uploading files using $_FILES. Demo: Build simple “Feedback Saver” app storing data in text file.
- Session 11 – Sessions & Cookies Topics to Cover: Setting and retrieving cookies. Creating PHP sessions with session_start(). Login/logout concept. Demo: User login using sessions and cookies. Depth: Discuss security practices - session hijacking prevention.
- Session 12 – Mini Project: “User Profile Manager” Objective: Apply all core concepts to build a mini CRUD-like app storing user info in arrays and session. Features: Add/Edit user. Display user list. Save session across pages.

Module 4) PHP- Advance PHP

- Session 1 – Introduction to OOP in PHP Topics to Cover: Procedural vs Object-Oriented PHP. Classes, Objects, Methods, and Properties. $this keyword and object instantiation. Demo: Create Student class with setDetails() and getDetails() methods. Trainer Focus: Highlight reusability and encapsulation benefits.
- Session 2 – Constructors & Destructors Topics to Cover: Magic methods: __construct(), __destruct(). Auto-initialization of properties. Constructor overloading via arguments. Demo: Create Product class auto-assigning name and price.
- Session 3 – Inheritance Topics to Cover: Extending classes using extends. Parent and child class relationships. Access modifiers: public, private, protected. Demo: Build Employee class inherited by Manager. Trainer Focus: Explain concept of DRY in inheritance chains.
- Session 4 – Polymorphism & Interfaces Topics to Cover: Method overriding. Interfaces and implements keyword. Abstract classes vs Interfaces. Demo: Shape ? Circle, Rectangle classes implementing calculateArea().
- Session 5 – Traits and Namespaces Topics to Cover: Traits for code reuse across classes. Namespaces for large project organization. Demo: Use trait Logger in multiple classes.
- Session 6 – Error & Exception Handling Topics to Cover: try, catch, finally, throw. Custom exception classes. Logging errors into file. Demo: Try-catch around file upload validation.
- Session 7 – PHP Forms with Validation Topics to Cover: Server-side validation using filter_var(). Sanitization of input data. Error messaging patterns. Demo: Build validated registration form. Depth: Compare client-side vs server-side validation.
- Session 8 – PHP File Upload System Topics to Cover: Upload directory management. Validating file size and type. Handling file rename conflicts. Demo: Create “Profile Picture Upload” feature.
- Session 9 – Working with MySQLi (Procedural Style) Topics to Cover: Connecting to DB, executing queries. mysqli_connect(), mysqli_query(), mysqli_fetch_assoc(). Demo: Display users table on webpage. Trainer Focus: Explain importance of error handling after queries.

- Session 10 – Working with PDO (Object-Oriented) Topics to Cover: PDO vs MySQLi. Prepared statements & parameter binding. Handling database exceptions. Demo: Login system with PDO prepared statements.
- Session 11 – Security in PHP Topics to Cover: SQL Injection, XSS, CSRF. Password hashing (password_hash(), password_verify()). Demo: Implement secure login with hashed password.
- Session 12 – MVC Architecture Fundamentals Topics to Cover: What is MVC (Model–View–Controller). Folder structure setup. Data flow between model, controller, view. Demo: Create simple MVC app manually: display user list. Depth: Diagram flow of request in MVC architecture.
- Session 13 – Building MVC Framework (Step 1) Topics to Cover: Core PHP routing. Create index.php as entry point. Controller invocation. Demo: Setup front controller routing system.
- Session 14 – Building MVC Framework (Step 2) Topics to Cover: Create Model class for database access. Create View for rendering HTML dynamically. Demo: Display data from database using MVC.
- Session 15 – CRUD with MVC Topics to Cover: Create, Read, Update, Delete operations. Model-Controller-View integration. Demo: “Employee Manager” CRUD in MVC structure.
- Session 16 – AJAX & jQuery with PHP Topics to Cover: Using AJAX to fetch and update data without page reload. Sending JSON data via $.ajax(). Demo: Live search in user table.
- Session 17 – Sending Emails via PHP Topics to Cover: PHP mail() vs PHPMailer. SMTP configuration. Demo: Send welcome mail to new users using PHPMailer.
- Session 18 – RESTful API Basics Topics to Cover: What is REST API. Sending/Receiving JSON response in PHP. CRUD endpoints via PHP. Demo: API returning list of students in JSON.
- Session 19 – Mini Project: “Employee Portal (MVC)” Project Objective: Develop Employee Management app using MVC: CRUD operations, Session-based login, AJAX search, Email notification.
- Session 20 – Code Refactoring & Review Topics to Cover: Improving readability and maintainability. Version control with Git. Demo: Refactor Controller and Model for reuse.

Module 5) PHP - Web services, API, Extensions - Industry

- Session 1 – RESTful API Fundamentals Topics to Cover: REST principles: resources, verbs, status codes. JSON structure & Postman usage. Demo: Build “Get All Products” PHP API returning JSON. AI Tip: Ask ChatGPT: “Generate standard REST API response schema.”
- Session 2 – Building PHP REST API (CRUD) Topics to Cover: Create endpoints for create, read, update, delete. HTTP methods: GET, POST, PUT, DELETE. Demo: CRUD API for Employee DB.
- Session 3 – Testing APIs with Postman Topics to Cover: Import/export requests. Postman collections, variables, and tests. Demo: Postman automated test for “Add Product.”
- Session 4 – Consuming External APIs Topics to Cover: Using file_get_contents() and cURL. Integrating weather/map/news APIs. Demo: Show live weather data via OpenWeatherMap API.
- Session 5 – Authentication & Tokens Topics to Cover: Basic Auth, API Key Auth, JWT. Secure endpoints. Demo: Implement JWT-based authentication for PHP API.
- Session 6 – Email & SMS API Integration Topics to Cover: SendGrid, Twilio, or Mailgun setup. Demo: Send OTP via Twilio SMS API. AI Add-on: Use ChatGPT to generate sample API documentation.

- Session 7 – Payment Gateway Integration Topics to Cover: Payment flow (frontend + backend). Setting up Razorpay / Stripe sandbox. Demo: Checkout page and capture payment via Razorpay API. Depth: Discuss webhook concept for payment confirmation.
- Session 8 – API Security & Error Handling Topics to Cover: Rate limiting, headers, HTTPS, status codes. Demo: Handle API error codes gracefully.
- Session 9 – Mini Project: “Smart Store API Suite” Goal: Create REST API for products, users, and orders (CRUD + authentication).
- Session 10 – Deployment & Documentation Topics to Cover: API documentation using Swagger or Postman Docs. Deploy API on free hosting. AI Integration: Ask ChatGPT to auto-generate API endpoint summary for documentation.

Module - 6) PHP - Laravel Framework

- Session 1 – Introduction to Laravel Topics to Cover: Why frameworks? Problems Laravel solves. MVC Recap (Model, View, Controller). Laravel installation via Composer. Folder structure overview (app, routes, config, resources). Demo: Install Laravel and run default project (php artisan serve). Trainer Focus: Explain role of artisan command-line tool.
- Session 2 – Routing in Laravel Topics to Cover: Basic routes (Route::get, Route::post). Route parameters and constraints. Named routes and route groups. Demo: Create routes for “Home,” “About,” and “Contact.” Depth: Emphasize RESTful design principles in routes.
- Session 3 – Controllers & Views Topics to Cover: Generating controllers with Artisan. Returning views with data. Passing data from controller to view. Demo: Create PagesController ? load Blade templates.
- Session 4 – Blade Templating Engine Topics to Cover: Blade syntax ({{ }}, @if, @foreach, @extends, @include). Template inheritance (master layouts). Demo: Create layout with header, footer, and content sections. Depth: Explain how Blade compiles into plain PHP.
- Session 5 – Request Handling Topics to Cover: Working with Request object. Handling GET, POST data. CSRF protection in forms. Demo: Form submission ? controller validation ? success message.
- Session 6 – Database Configuration Topics to Cover: Setting up .env for DB connection. Configuring MySQL credentials. Running migrations (php artisan migrate). Demo: Migrate default users table.
- Session 7 – Migrations & Schema Builder Topics to Cover: Creating custom migrations. Table fields, foreign keys, rollback. Demo: Create products table with migration file.
- Session 8 – Seeding & Factories Topics to Cover: Database seeding using Faker. Model factories for fake data. Demo: Seed 20 sample product records.
- Session 9 – Eloquent ORM (Part 1) Topics to Cover: Models, naming conventions. CRUD using Eloquent: create(), find(), update(), delete(). Demo: Product CRUD using Eloquent. Depth: Show SQL queries behind Eloquent via Laravel Debugbar.
- Session 10 – Eloquent ORM (Part 2) Topics to Cover: Relationships: One-to-One, One-to-Many, Many-to-Many. Eager loading and lazy loading. Demo: Define Category–Product relationship.
- Session 11 – Validation Topics to Cover: Built-in validation rules. Custom messages and error display. Form Request validation class. Demo: Form validation for “Add Product.”
- Session 12 – File Uploads Topics to Cover: Handling file uploads using Request object. Validating file type, size, and naming. Storing images in public/storage. Demo: Add product image upload feature.

- Session 13 – Authentication (Part 1) Topics to Cover: Laravel Breeze / UI package for authentication. Registration and login scaffolding. Demo: Install Breeze, demonstrate login/register flow. Trainer Focus: Explain middleware protection and session handling.
- Session 14 – Authentication (Part 2) Topics to Cover: Middleware concepts (auth, guest). Customizing Auth views. Logout and password reset. Demo: Protect product CRUD routes with auth middleware.
- Session 15 – Eloquent Advanced (Query Optimization) Topics to Cover: Query Scopes, Accessors, and Mutators. Raw queries and query builder. Demo: Global scope for filtering active products.
- Session 16 – Mail & Notifications Topics to Cover: Configuring mail via Gmail SMTP. Sending Mailable classes and notifications. Demo: Send order confirmation email to user.
- Session 17 – Events, Listeners & Queues Topics to Cover: Creating Events & Listeners. Queue basics and queue:work. Demo: Send async emails using queue. Depth: Explain use cases for background jobs (order confirmations, report generation).
- Session 18 – Middleware & Security Topics to Cover: Creating custom middleware. CSRF & XSS protection. Role-based access control (admin/user). Demo: Custom middleware that blocks inactive users.
- Session 19 – API Development with Laravel (Part 1) Topics to Cover: Creating RESTful controllers. JSON response handling. Demo: API for Product CRUD (returning JSON). AI Integration: Use Postman AI Assistant to auto-test endpoints.
- Session 20 – API Development with Laravel (Part 2) Topics to Cover: Authentication using Sanctum. API Token management. Demo: Login and fetch data via token-based API.
- Session 21 – AJAX & Frontend Integration Topics to Cover: jQuery AJAX calls to Laravel routes. Dynamic updates without page reload. Demo: Live product search via AJAX and JSON.
- Session 22 – Performance Optimization Topics to Cover: Cache, Config, and Route optimization. Using php artisan optimize. Lazy vs eager loading performance test. Demo: Profile queries with Laravel Debugbar.
- Session 23 – Deployment & Environment Management Topics to Cover: Hosting Laravel on shared server (cpanel). Using .env for config. Migration and production DB sync. Demo: Deploy Laravel app to Hostinger / InfinityFree.
- Session 24 – Mini Project: “E-Commerce Portal (Laravel)” Project Features: Product CRUD with images. User registration/login. Add to cart, checkout, and order confirmation mail. AI Integration: Copilot for route generation, ChatGPT for email templates.
- Session 25 – Code Review & Project Documentation Topics to Cover: Clean code and naming conventions. Generating project documentation. GitHub upload and version tracking. Demo: Push final project to GitHub with README.md.

Module - 7) AI Tools for PHP Developers

- Session 1 – AI in Web Development: Concepts & Tools Topics to Cover: Understanding AI-assisted development. Overview: ChatGPT, GitHub Copilot, Postman AI, Firefly, Leonardo, and OpenAI APIs. Use cases: code generation, debugging, design, content creation. Trainer Focus: Show practical examples where AI tools save developer time. Demo: Ask ChatGPT: “Explain Laravel route groups in simple terms with code examples.” Discussion: Compare human-written vs AI-written snippets for clarity and

- Session 2 – Using ChatGPT for PHP & Laravel Productivity Topics to Cover: Prompting strategies for code assistance. Using ChatGPT to: Generate CRUD code. Debug PHP warnings. Refactor old procedural code into OOP. Demo: Provide buggy code to ChatGPT and correct it live. Depth: Explain that AI outputs are assistive, not authoritative - always test before deploying.
- Session 3 – GitHub Copilot in VS Code Topics to Cover: Installation and activation of Copilot. How Copilot understands comments ? code generation. Real-time suggestion workflow in Laravel controllers. Demo: Type a controller comment (e.g., "// create API endpoint for fetching user orders") ? watch Copilot auto-generate code. Trainer Tip: Encourage learners to use Copilot for repetitive functions (CRUD, migrations).
- Session 4 – AI for Design & Content Topics to Cover: Figma AI / Uizard for design mockups. Adobe Firefly & Leonardo.ai for banners and visuals. ChatGPT for SEO content (meta tags, product descriptions). Demo: Generate hero image + tagline for Laravel e-commerce homepage using Firefly. AI Add-on: Ask ChatGPT: "Write SEO title, meta description, and alt text for a Laravel e-commerce site."
- Session 5 – Integrating OpenAI API with Laravel Topics to Cover: Register for OpenAI API key. Install openai-php/client package. Make API call from Laravel controller. Process and display AI-generated response. Demo: Create "AI Text Generator" that summarizes product descriptions. Trainer Focus: Explain API call structure, key handling via .env.
- Session 6 – Mini Project: "AI Product Description Generator" Project Objective: Integrate Laravel + OpenAI API to generate SEO-optimized product descriptions. Features: User inputs product name and features. Laravel controller calls OpenAI API. AI-generated text displayed in formatted HTML. Demo Flow: Create OpenAI service class. Controller handles user request. Response shown on Blade page.